Linear Models Chapter 5

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Chapter 5

Linear models

5.1 Relationship among variables

We wish to examine the relationships linking the variables under consideration.

- There is a response variable or dependent variable or simply a left-hand-side variable

$$\begin{array}{ccc} Y & = \ldots & \\ \uparrow & & \uparrow \\ left & & right \end{array}$$

- On the right hand side there are explanatory variables, or covariates, sometimes also called independent variables: X_1, X_2, ..., X_p.

- The right hand side is a sum of two terms

$$Y = r(X_1, X_2, \ldots, X_p) + \varepsilon \tag{5.1}$$

  The first term is the systematic component. The term ε is called the error term, or disturbance term, which is, preferably, free from any connection to the explanatory variables and has zero expectation $\mathrm{E}(\varepsilon) = 0$.

- Equation (5.1) is said to define a regression model, since $\mathrm{E}(Y \mid X_i = x_i,\ i = 1, \ldots, p) = r(x_1, \ldots, x_p)$. However, we should specify the function $r(\cdot)$ and define the (stochastic) property of ε before we can say we model the stochastic properties of y.

In particular we consider the case where $r(\cdot)$ is a linear function (of the parameters)

$$Y = X_1\beta_1 + \ldots + X_p\beta_p + \varepsilon \tag{5.2}$$

Notice that the model

$$Y = X_1\beta_1 + X_2^2\beta_2 + \varepsilon \tag{5.3}$$

is linear as well (not in the variables X_1 and X_2, but in the parameters).

Sometimes we have to apply a transformation. If

$$Z = e^{\sum_{j=1}^{p} X_j \beta_j} e^{\varepsilon}, \tag{5.4}$$

then

$$Y = \log(Z) = \sum_{j=1}^{p} X_j \beta_j + \varepsilon. \tag{5.5}$$

The error term ε is used to capture cases where Y depends approximately linearly on the explanatory variables $X_1, \ldots, X_p$. We would like it to be "small".

Example 1
Treatment effect
Suppose we divide a group of n subjects at random in two subgroups with n_1 and n_2 observations. The first group is the treatment group who receives a treatment, such as an injection with some substance, or a training.

The other subgroup, control group, doesn't get any treatment but in all other aspects the control group is comparable to the treatment group since the selection was random.

Let $y_1, \ldots, y_n$ be the measurements of a property that might be affected by the treatment.

We can model this as follows

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_{i,} \tag{5.6}$$

where $x_{i1} = 1$ for all $i = 1, \ldots, n$, and $x_{i2} = 1$ if subject i is in the treatment group and $x_{i2} = 0$ if i is in the control group, $i = 1, \ldots, n$.

We have the following situation

- *We observe the x variables: we know who gets the treatment.*
- *We also observe the dependent variable: we measure it.*
- *As the measurements $y_1, \ldots, y_n$ will vary within the groups, the values of ε_i will vary within the groups.*
- *We don't observe the disturbances and we don't know the parameters β_1 and β_2*
- *The goal of the analysis is to make inferences about β_1 and β_2.*
- *The constant term, β_1, could be interpreted as an average level of the y variable without a treatment. The average level of ε could be taken zero since if it were not zero it could be subtracted from ε and added to β_1 without affecting the equality* (5.6).
- *The slope coefficient β_2 is the treatment effect: without treatment the average level is β_1, and with treatment the level is $\beta_1 + \beta_2$.*

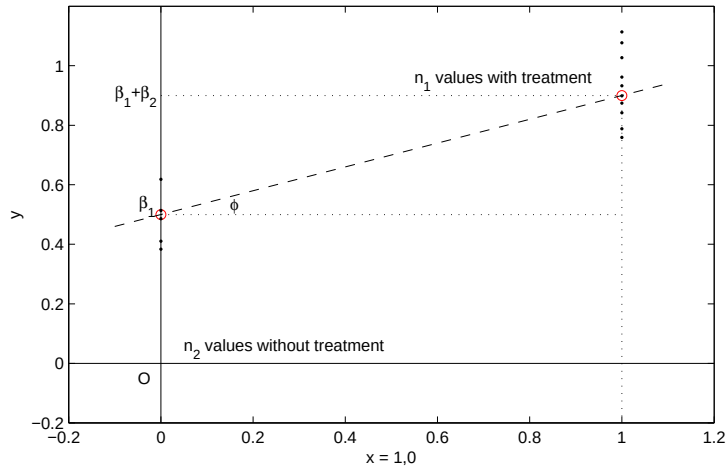

Figure 5.1: Treatment effect

$$\text{regression coefficients} \begin{cases} \beta_1 : \text{intercept, constant term} \\ \beta_2 = \tan(\phi), \text{slope coefficient} \end{cases}$$

The regression coefficients are unknown parameters. We assume they are constants that do not depend on the particular sample of n subjects.

We could use the sample (y_i, x_{i2}), $i = 1, \ldots, n$, to estimate the parameters β_1 and β_2.

That is, computing statistics $\hat{\beta}_1$ and $\hat{\beta}_2$, which can be calculated based on the data (y_i, x_{i2}), $i = 1, \ldots, n$, that might be close to β_1 and β_2, respectively, in some way.

For example, a natural way to estimate in the treatment example would be to let $\hat{\beta}_1$ be the average of y values over the control group subjects, $\hat{\beta}_1 = (1/n_2) \sum_{i=n_1+1}^{n} y_i$, and to let $\hat{\beta}_2$ be the average of y over the treatment group, where we subtract $\hat{\beta}_1$, i.e. $\hat{\beta}_2 = (1/n_1) \sum_{i=1}^{n_1} y_i - \hat{\beta}_1$.

Of course, if we would have taken another sample of n subjects, or if we had used another random devision into two subgroups, we would have found, using the same estimators (ways to compute estimates), other estimates (outcomes).

So to define "closeness" between the estimators $\hat{\boldsymbol{\beta}}_i$ and the parameters $\boldsymbol{\beta}_i$, $i = 1, 2$, we have to specify a stochastic model for the errors ε_i that determines the multivariate distribution for our estimators.

5.2 Least squares

5.2.1 Ordinary least square (OLS)

Before we specify a stochastic model, we use an approach to compute estimates $\hat{\boldsymbol{\beta}}$ based on the assumption that the disturbances are "small" if true parameter values are used, even

though they are unknown. Let

$$y_i = \sum_{j=1}^{p} \beta_j x_{ij} + \varepsilon_i. \tag{5.7}$$

The method of least squares solves the optimization problem

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}=(\beta_1,\ldots,\beta_p)'}{\arg\min} \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2. \tag{5.8}$$

The solution is found by solving

$$\frac{\partial \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2}{\partial \beta_k} = 0, \tag{5.9}$$

for $k = 1, \ldots, p$, which amounts to solving

$$\sum_{i=1}^{n} 2 \left(y_i - \sum_{j=1}^{p} \beta_j x_{ij} \right) (-x_{ik}) = 0, \tag{5.10}$$

for $k = 1, \ldots, p$. It seems hard to solve if p is larger than one.

Remark 1
Simple regression
For $p = 1$ the minimization problem amounts to finding

$$\hat{\beta} = \underset{\beta}{\arg\,\text{solve}} \sum_{i=1}^{n} (y_i - x_i\beta) x_i = 0. \tag{5.11}$$

We find quite easily

$$\hat{\beta} = \frac{\sum_{i=1}^{n} y_i x_i}{\sum_{i=1}^{n} x_i^2}. \tag{5.12}$$

When $p = 2$ and $y_i = \beta_1 + x_i\beta_2$, so that the first explanatory variable has a constant value $x_{i1} = 1$ and β_1 is called the constant term*, and $x_i = x_{i2}$, we talk about* simple regression*, or bivariate regresssion, since there are only two variables involved. In that case the solution of* (5.10) *is also relatively easy to formulate.*

The solution should satisfy two equations, which are given by

$$\sum_{i=1}^{n} (y_i - \hat{\beta}_1 - x_i\hat{\beta}_2) = 0, \qquad \textit{and} \qquad \sum_{i=1}^{n} (y_i - \hat{\beta}_1 - x_i\hat{\beta}_2) x_i = 0. \tag{5.13}$$

Consequently, using $\overline{x}_n = \sum_{i=1}^n x_i/n$ *and combining the two equations, the solution should satisfy*

$$\sum_{i=1}^n (y_i - \hat{\beta}_1 - x_i\hat{\beta}_2)(x_i - \overline{x}_n) = 0. \tag{5.14}$$

Now we can apply a simple rule.

Simple rule*:*

$\boxed{\text{If } \sum_{i=1}^n a_i = 0, \text{ then for any constant } c \text{ we have } \sum_{i=1}^n w_i a_i = \sum_{i=1}^n (w_i - c)a_i.}$

Using this rule we find, based on (5.14)

$$\sum_{i=1}^n (y_i - x_i\hat{\beta}_2)(x_i - \overline{x}_n) = 0. \tag{5.15}$$

So,

$$\hat{\beta}_2 = \frac{\sum_{i=1}^n y_i(x_i - \overline{x}_n)}{\sum_{i=1}^n x_i(x_i - \overline{x}_n)} \tag{5.16}$$

$$= \frac{\sum_{i=1}^n (y_i - \overline{y}_n)(x_i - \overline{x}_n)}{\sum_{i=1}^n (x_i - \overline{x}_n)^2} = \frac{s_{xy}}{s_x^2}, \tag{5.17}$$

where s_{xy} *is the sample covariance of* x *and* y *and* s_x^2 *is the sample variance of* x*. Futhermore, using the first equation in* (5.13)*, we find*

$$\hat{\beta}_1 = \overline{y}_n - \overline{x}_n\hat{\beta}_2. \tag{5.18}$$

This describes the least squares solution $\hat{\beta}_1$ *and* $\hat{\beta}_2$ *for simple regression. We return to this case in Section 5.2.3.*

To continue with our problem of finding the least squares solution for general p, we turn to matrix algebra. In matrix notation the equation (5.7) is given by

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{5.19}$$

where $\boldsymbol{y} = (y_1, \ldots, y_n)'$ and $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_n)'$ are $n \times 1$ vectors, $\boldsymbol{X} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_p)$ is an $n \times p$ matrix of full column rank with elements $\{x_{ij}\}$, and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)'$. Now least squares amounts to minimize $\|\hat{\boldsymbol{\varepsilon}}\| = \|\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}\|$, or

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} \tag{5.20}$$

where

$$\boldsymbol{\varepsilon} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}. \tag{5.21}$$

Now

$$\frac{\partial(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})}{\partial\boldsymbol{\beta}} = 2\left\{\frac{\partial(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})}{\partial\boldsymbol{\beta}'}\right\}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) \tag{5.22}$$

$$= 2\left\{-\boldsymbol{X}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})\right\} \tag{5.23}$$

$$= -2\boldsymbol{X}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}). \tag{5.24}$$

Alternatively,

$$\frac{\partial(\boldsymbol{y}-\boldsymbol{X\beta})'(\boldsymbol{y}-\boldsymbol{X\beta})}{\partial\boldsymbol{\beta}} = \frac{\partial(\boldsymbol{y}'\boldsymbol{y}-\boldsymbol{y}'\boldsymbol{X\beta}-\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{y}+\boldsymbol{\beta X}'\boldsymbol{X\beta})}{\partial\boldsymbol{\beta}} \tag{5.25}$$

$$= \frac{\partial(\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X\beta}-2\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{y})}{\partial\boldsymbol{\beta}} \tag{5.26}$$

$$= 2\boldsymbol{X}'\boldsymbol{X\beta}-2\boldsymbol{X}'\boldsymbol{y}, \tag{5.27}$$

and the normal equations are given by

$$2\boldsymbol{X}'\boldsymbol{X}\hat{\boldsymbol{\beta}}-2\boldsymbol{X}'\boldsymbol{y}=\boldsymbol{0}. \tag{5.28}$$

The OLS solution is given by

$$\hat{\boldsymbol{\beta}}=(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}, \tag{5.29}$$

which is possible to compute if $\boldsymbol{X}$ has f.c.r., so $\boldsymbol{X}'\boldsymbol{X}$ is nonsingular.

5.2.2 Projections

We could have found the OLS solution also by using projection matrices. Let

$$\boldsymbol{P_X}=\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \tag{5.30}$$

then both $\boldsymbol{P_X}$ and $\boldsymbol{I}_n-\boldsymbol{P_X}$ are projection matrices, and

$$(\boldsymbol{y}-\boldsymbol{X\beta})'(\boldsymbol{y}-\boldsymbol{X\beta}) = (\boldsymbol{y}-\boldsymbol{X\beta})'\boldsymbol{P_X}(\boldsymbol{y}-\boldsymbol{X\beta})+(\boldsymbol{y}-\boldsymbol{X\beta})'(\boldsymbol{I}_n-\boldsymbol{P_X})(\boldsymbol{y}-\boldsymbol{X\beta}) \tag{5.31}$$

$$= (\boldsymbol{y}-\boldsymbol{X\beta})'\boldsymbol{P_X}(\boldsymbol{y}-\boldsymbol{X\beta})+\boldsymbol{y}'(\boldsymbol{I}_n-\boldsymbol{P_X})\boldsymbol{y} \tag{5.32}$$

where $\boldsymbol{y}'(\boldsymbol{I}_n-\boldsymbol{P_X})\boldsymbol{y}$ does not depend on $\boldsymbol{\beta}$, and the first term on the right hand side is nonnegative, since $\boldsymbol{P_X}\geq 0$. We find for $\hat{\boldsymbol{\beta}}$ satisfying $\boldsymbol{X}'(\boldsymbol{y}-\boldsymbol{X}\hat{\boldsymbol{\beta}})=0$ the first term is minimal, i.e. it equals zero for this choice of $\hat{\boldsymbol{\beta}}$. So, we find again the OLS solution

$$\hat{\boldsymbol{\beta}}=(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}. \tag{5.33}$$

Furthermore, we find

$$\boldsymbol{X}\hat{\boldsymbol{\beta}}=\boldsymbol{P_X}\boldsymbol{y}, \tag{5.34}$$

$$\hat{\boldsymbol{\varepsilon}}=\boldsymbol{y}-\boldsymbol{X}\hat{\boldsymbol{\beta}}=\boldsymbol{y}-\boldsymbol{P_X}\boldsymbol{y}=(\boldsymbol{I}_n-\boldsymbol{P_X})\boldsymbol{y}. \tag{5.35}$$

So, the ordinary least square (OLS) solution decomposes the dependent vector $\boldsymbol{y}$ into two orthogonal vectors:

$$\boldsymbol{y}=\boldsymbol{P_X}\boldsymbol{y}+(\boldsymbol{I}_n-\boldsymbol{P_X})\boldsymbol{y} \tag{5.36}$$

$$=\boldsymbol{X}\hat{\boldsymbol{\beta}}+\hat{\boldsymbol{\varepsilon}}, \tag{5.37}$$

where $\boldsymbol{X}'\hat{\boldsymbol{\varepsilon}}=\boldsymbol{0}$, and

$$\|\boldsymbol{y}\|^2=\|\boldsymbol{X}\hat{\boldsymbol{\beta}}\|^2+\|\hat{\boldsymbol{\varepsilon}}\|^2. \tag{5.38}$$

Example 2
Treatment effect revisited

Let's see how OLS works out for the treatment effect in Example 1:

$$\boldsymbol{y}_i = \beta_1 + x_{i2}\beta_2 + \boldsymbol{\varepsilon}_i, \quad i = 1, \ldots, n, \tag{5.39}$$

where x_{i2} is binary or dichotomous, having only two values: 0 or 1.

We could arrange the n observations into the first n_1 observations of subjects who did receive the treatment, $x_{i2} = 1$, $i = 1, \ldots, n_1$, and the last n_2 who did not get the treatment, $x_{i2} = 0$, $i = n_{1+1}, \ldots, n$:

$$\boldsymbol{y} = \begin{pmatrix} \boldsymbol{y}_1 \\ \boldsymbol{y}_2 \end{pmatrix} = \begin{pmatrix} \boldsymbol{\iota}_{n_1} \\ \boldsymbol{\iota}_{n_2} \end{pmatrix} \beta_1 + \begin{pmatrix} \boldsymbol{\iota}_{n_1} \\ \boldsymbol{0} \end{pmatrix} \beta_2 + \begin{pmatrix} \boldsymbol{\varepsilon}_1 \\ \boldsymbol{\varepsilon}_2 \end{pmatrix} \tag{5.40}$$

$$= \boldsymbol{\iota}_n \beta_1 + \boldsymbol{x}_2 \beta_2 + \boldsymbol{\varepsilon} \tag{5.41}$$

$$= \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}. \tag{5.42}$$

where $\boldsymbol{y}$, $\boldsymbol{x}_1$, $\boldsymbol{\varepsilon}$ and $\boldsymbol{X}$ and $\boldsymbol{\beta}$ have been implicitly defined.

The OLS solution is given by

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}. \tag{5.43}$$

To look more closely at the solution, we should study the inverse of the partitioned matrix

$$\boldsymbol{X}'\boldsymbol{X} = (\boldsymbol{\iota}_n, \boldsymbol{x}_2)'(\boldsymbol{\iota}_n, \boldsymbol{x}_2) \tag{5.44}$$

$$= \begin{pmatrix} \boldsymbol{\iota}_n'\boldsymbol{\iota}_n & \boldsymbol{\iota}_n'\boldsymbol{x}_2 \\ \boldsymbol{x}_2'\boldsymbol{\iota}_n & \boldsymbol{x}_2'\boldsymbol{x}_2 \end{pmatrix} \tag{5.45}$$

$$= \begin{pmatrix} n & n_1 \\ n_1 & n_1 \end{pmatrix}, \tag{5.46}$$

$$(\boldsymbol{X}'\boldsymbol{X})^{-1} = \frac{1}{nn_1 - n_1^2} \begin{pmatrix} n_1 & -n_1 \\ -n_1 & n \end{pmatrix} = \begin{pmatrix} n_2^{-1} & -n_2^{-1} \\ -n_2^{-1} & n_1^{-1} + n_2^{-1} \end{pmatrix}. \tag{5.47}$$

Furthermore,

$$\boldsymbol{X}'\boldsymbol{y} = (\boldsymbol{\iota}_n, \boldsymbol{x}_2)' \begin{pmatrix} \boldsymbol{y}_1 \\ \boldsymbol{y}_2 \end{pmatrix} = \begin{pmatrix} \boldsymbol{\iota}_{n_1}'\boldsymbol{y}_1 + \boldsymbol{\iota}_{n_2}'\boldsymbol{y}_2 \\ \boldsymbol{\iota}_{n_1}'\boldsymbol{y}_1 \end{pmatrix}. \tag{5.48}$$

So,

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} = \begin{pmatrix} \frac{\boldsymbol{\iota}_{n_2}'\boldsymbol{y}_2}{n_2} \\ \frac{\boldsymbol{\iota}_{n_1}'\boldsymbol{y}_1}{n_1} - \frac{\boldsymbol{\iota}_{n_2}'\boldsymbol{y}_2}{n_2} \end{pmatrix} = \begin{pmatrix} \bar{y}_2 \\ \bar{y}_1 - \bar{y}_2 \end{pmatrix}, \tag{5.49}$$

and $\hat{\beta}_1 = \bar{y}_2$, and $\hat{\beta}_2 = \bar{y}_1 - \bar{y}_2$. This is the same solution as the "natural solution" found earlier.

Alternatively, we could have found the OLS solution in the treatment-effect example also by considering the partitioned vectors

$$\boldsymbol{y}_1 = \boldsymbol{\iota}_{n_1}(\beta_1 + \beta_2) + \boldsymbol{\varepsilon}_1, \tag{5.50}$$
$$\boldsymbol{y}_2 = \boldsymbol{\iota}_{n_2}\beta_1 + \boldsymbol{\varepsilon}_2. \tag{5.51}$$

We know $\|\boldsymbol{\varepsilon}_2\|$ *is minimized by* $\hat{\beta}_1$ *given by*

$$\hat{\beta}_1 = (\boldsymbol{\iota}_{n_2}'\boldsymbol{\iota}_{n_2})^{-1}\boldsymbol{\iota}_{n_2}'\boldsymbol{y}_2 \tag{5.52}$$
$$= \frac{\boldsymbol{\iota}_{n_2}'\boldsymbol{y}_2}{n_2} = \overline{y}_2. \tag{5.53}$$

Given any value of $\hat{\beta}_1$ *we can minimize* $\|\boldsymbol{\varepsilon}_1\|$ *by choosing* β_2 *optimally. We have*

$$\hat{\beta}_1 + \hat{\beta}_2 = (\boldsymbol{\iota}_{n_1}'\boldsymbol{\iota}_{n_1})^{-1}\boldsymbol{\iota}_{n_1}'\boldsymbol{y}_1 = \overline{y}_1, \tag{5.54}$$

which agrees with the OLS solution found above.

5.2.3 Simple linear regression

Check Example 5.2.4. in the book

To describe simple regression, as discussed in Remark 1, let

$$\boldsymbol{X} = (\boldsymbol{\iota}_n, \boldsymbol{x}) = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \tag{5.55}$$

so

$$\boldsymbol{y} = \boldsymbol{\iota}_n\beta_1 + \boldsymbol{x}\beta_2 + \boldsymbol{\varepsilon} \tag{5.56}$$
$$= \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}. \tag{5.57}$$

We have $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$, and

$$\boldsymbol{X}'\boldsymbol{X} = \begin{pmatrix} n & \sum_i x_i \\ \sum x_i & \sum_i x_i^2 \end{pmatrix}, \tag{5.58}$$

$$(\boldsymbol{X}'\boldsymbol{X})^{-1} = \frac{1}{n\sum_i x_i^2 - (\sum_i x_i)^2}\begin{pmatrix} \sum_i x_i^2 & -\sum_i x_i \\ -\sum_i x_i & n \end{pmatrix}, \tag{5.59}$$

$$\boldsymbol{X}'\boldsymbol{y} = \begin{pmatrix} \sum_i y_i \\ \sum_i x_i y_i \end{pmatrix}. \tag{5.60}$$

So,

$$\hat{\boldsymbol{\beta}} = \frac{1}{n\sum_i x_i^2 - (\sum_i x_i)^2}\begin{pmatrix} \sum_i y_i \sum_i x_i^2 - \sum_i x_i \sum_i x_i y_i \\ n\sum_i x_i y_i - \sum_i x_i \sum_i y_i \end{pmatrix}. \tag{5.61}$$

Define the sample means

$$\bar{x}_n = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \bar{y}_n = \frac{1}{n}\sum_{i=1}^{n} y_i, \tag{5.62}$$

the sample variances

$$s_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \tag{5.63}$$

$$s_y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2, \tag{5.64}$$

the sample covariance

$$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}), \tag{5.65}$$

and the sample correlation

$$r = \frac{s_{xy}}{s_x s_y}. \tag{5.66}$$

As

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 = \left(\sum_{i=1}^{n} x_i^2\right) - 2\bar{x}\left(\sum_{i=1}^{n} x_i\right) + n\bar{x}^2 \tag{5.67}$$

$$= \left(\sum_{i=1}^{n} x_i^2\right) - n\bar{x}^2 \tag{5.68}$$

$$= \left(\sum_{i=1}^{n} x_i^2\right) - \frac{(\sum_{i=1}^{n} x_i)^2}{n} \tag{5.69}$$

and

$$\sum_{i=1}^{n}(x_i - \bar{x})(y - \bar{y}) = \left(\sum_{i=1}^{n} x_i y_i\right) - \frac{(\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n}, \tag{5.70}$$

we find for the OLS solution:

$$\hat{\boldsymbol{\beta}} = \frac{1}{n(\sum_i x_i^2) - (\sum_i x_i)^2}\begin{pmatrix} \sum_i y_i \sum_i x_i^2 - \sum_i x_i \sum_i x_i y_i \\ n\sum_i x_i y_i - \sum_i x_i \sum_i y_i \end{pmatrix} = \begin{pmatrix} \bar{y} - \frac{s_{xy}}{s_x^2}\bar{x} \\ \frac{s_{xy}}{s_x^2} \end{pmatrix}. \tag{5.71}$$

Indeed, if

$$\boldsymbol{y} = \boldsymbol{\iota}_n \hat{\beta}_1 + \boldsymbol{x}\hat{\beta}_2 + \hat{\boldsymbol{\varepsilon}}, \tag{5.72}$$

and

$$\boldsymbol{X}'\hat{\boldsymbol{\varepsilon}} = \begin{pmatrix} \boldsymbol{\iota}_n'\hat{\boldsymbol{\varepsilon}} \\ \boldsymbol{x}'\hat{\boldsymbol{\varepsilon}} \end{pmatrix} = \boldsymbol{0}, \tag{5.73}$$

then

$$\bar{y} = \frac{\boldsymbol{\iota}_n' \boldsymbol{y}}{n} = \frac{\boldsymbol{\iota}_n'(\boldsymbol{\iota}_n \hat{\beta}_1 + \boldsymbol{x}\hat{\beta}_2 + \hat{\boldsymbol{\varepsilon}})}{n} \tag{5.74}$$

$$= \hat{\beta}_1 + \bar{x}\hat{\beta}_2, \tag{5.75}$$

and

$$(\boldsymbol{x} - \boldsymbol{\iota}\bar{x}_n)'\boldsymbol{y} = (\boldsymbol{x} - \boldsymbol{\iota}\bar{x}_n)'\boldsymbol{x}\hat{\beta}_2, \tag{5.76}$$

So

$$\hat{\beta}_1 = \bar{y} - \bar{x}\hat{\beta}_2, \tag{5.77}$$

$$\hat{\beta}_2 = \frac{(\boldsymbol{x} - \boldsymbol{\iota}\bar{x}_n)'\boldsymbol{y}}{(\boldsymbol{x} - \boldsymbol{\iota}\bar{x}_n)'\boldsymbol{x}} \tag{5.78}$$

$$= \frac{(\boldsymbol{x} - \boldsymbol{\iota}\bar{x}_n)'(\boldsymbol{y} - \boldsymbol{\iota}\bar{y}_n)}{(\boldsymbol{x} - \boldsymbol{\iota}\bar{x}_n)'(\boldsymbol{x} - \boldsymbol{\iota}\bar{x}_n)} = \frac{s_{xy}}{s_x^2}. \tag{5.79}$$

In particular, we find for the OLS disturbances $\hat{\boldsymbol{\varepsilon}}$ that $\boldsymbol{X}'\hat{\boldsymbol{\varepsilon}} = \boldsymbol{0}$. This implies the sample mean of the OLS disturbance is zero, $\overline{\hat{\boldsymbol{\varepsilon}}_n} = \boldsymbol{0}$, and the sample covariance of the x_i's and $\hat{\varepsilon}_i$'s is zero as well

$$s_{x\hat{\varepsilon}} = (n-1)^{-1} \sum_{i=1}^{n} (x_i - \bar{x})(\hat{\varepsilon}_i - \overline{\hat{\boldsymbol{\varepsilon}}}) \tag{5.80}$$

$$= (n-1)^{-1} \sum_{i=1}^{n} x_i(\hat{\varepsilon}_i - \overline{\hat{\boldsymbol{\varepsilon}}}) \tag{5.81}$$

$$= (n-1)^{-1} \sum_{i=1}^{n} x_i\hat{\varepsilon}_i = \frac{\boldsymbol{x}'\hat{\boldsymbol{\varepsilon}}}{n-1} = 0. \tag{5.82}$$

So, the sample correlation between the x_i's and $\hat{\boldsymbol{\varepsilon}}_i$'s is zero.

The OLS regression line is given by by points $(\hat{y}, x)$, where $\hat{y} = \hat{\beta}_1 + x\hat{\beta}_2$. The sample mean $(\bar{y}, \bar{x})$ is located on the OLS regression line since

$$\bar{y} = \hat{\beta}_1 + \hat{\beta}_2\bar{x} \tag{5.83}$$

Finally, as

$$(n-1)s_y^2 = \sum_{i=1}^{n} (y_i - \bar{y})^2 \tag{5.84}$$

$$= \sum_{i=1}^{n} (y_i - \bar{y}) \left(\hat{\beta}_1 + x_i\hat{\beta}_2 + \hat{\varepsilon}_i - (\hat{\beta}_1 + \bar{x}\hat{\beta}_2) \right) \tag{5.85}$$

$$= \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})\hat{\beta}_2 + \sum_{i=1}^{n} (y_i - \bar{y})\hat{\varepsilon}_i \tag{5.86}$$

$$= (n-1)s_{xy}\hat{\beta}_2 + \sum_{i=1}^{n} \hat{\varepsilon}_i^2 \tag{5.87}$$

$$= (n-1)\frac{s_{xy}^2}{s_x^2} + (n-1)s_{\hat{\varepsilon}}^2 = (n-1)\left(\frac{s_{xy}^2}{s_x^2 s_y^2} s_y^2 + s_{\hat{\varepsilon}}^2 \right) \tag{5.88}$$

we find

$$s_y^2 = r^2 s_y^2 + s_{\hat{\varepsilon}}^2 \tag{5.89}$$

A proportion r^2 of the sample variance s_y^2 is explained, and the residuals $\hat{\boldsymbol{\varepsilon}}$ contain a proportion

$$\frac{s_{\hat{\varepsilon}}^2}{s_y^2} = 1 - r^2 \tag{5.90}$$

unexplained sample variance.

5.3 The classical stochastic linear model

(Subsections of Section 5.2 in the book)

5.3.1 The model

Assume the data y_i, $i = 1, \ldots, n$, have been generated by the classic stochastic linear model given by

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \epsilon, \tag{5.91}$$

where

(i) $\boldsymbol{X} : n \times p$, f.c.r. nonrandom;

(ii) $\mathrm{E}(\boldsymbol{\varepsilon}) = \boldsymbol{0}$;

(iii) $\mathrm{Var}(\boldsymbol{\varepsilon}) = \mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}') = \sigma^2 \boldsymbol{I}_n$.

That is to say, the matrix $\boldsymbol{X}$ is fixed, $\boldsymbol{\beta}$ is an unknown parameter vector, $\boldsymbol{\varepsilon}$ is random with zero expectation, the variances of the disturbances $\boldsymbol{\varepsilon}_i$ are all equal to σ^2, which is also an unknown parameter, and the disturbances are uncorrelated: $\mathrm{E}(\boldsymbol{\varepsilon}_i \boldsymbol{\varepsilon}_j) = 0$ if $i \neq j$.

Remark 2
Randomness of $\boldsymbol{X}$.
In many cases explanatory variables can be considered as random just as the dependent variable is. In that case we may condition on $\boldsymbol{X}$:

(ii') $\mathrm{E}(\boldsymbol{\varepsilon} \mid \boldsymbol{X}) = \boldsymbol{0}$;

(iii') $\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \boldsymbol{X}) = \sigma^2 \boldsymbol{I}_n$.

These alternative assumptions (ii') *and* (iii') *could then be used instead of* (ii) *and* (iii).

In that case the results for the classical model hold conditional on the given value of $\boldsymbol{X}$ in the sample. So, for example, we would not have

$$\mathrm{E}(\boldsymbol{X}'\boldsymbol{\varepsilon}) = \boldsymbol{X}' \mathrm{E}(\boldsymbol{\varepsilon}),$$

since $\boldsymbol{X}$ is random, but if (ii') *holds, then*

$$\mathrm{E}(\boldsymbol{X}'\boldsymbol{\varepsilon} \mid \boldsymbol{X}) = \boldsymbol{X}' \mathrm{E}(\boldsymbol{\varepsilon} \mid \boldsymbol{X}) = 0 \tag{5.92}$$

does holds true.

In the sequel we assume $\boldsymbol{X}$ is not random—we know the results could be easily generalized.

Remark 3
Errors in variables
If $\boldsymbol{X}$ is random we may use the alternative assumption $\mathrm{E}(\boldsymbol{\varepsilon} \mid \boldsymbol{X}) = 0$ and $\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \boldsymbol{X}) = \sigma^2 \boldsymbol{I}_n$. However, there are important cases where these assumptions cannot hold. Consider, for example, the case of error in variables, where the explanatory variables contain measurement error. Let, the model be given by

$$\boldsymbol{y} = \boldsymbol{x}_* \boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{5.93}$$
$$\boldsymbol{x} = \boldsymbol{x}_* + \boldsymbol{\eta}, \tag{5.94}$$

where $\boldsymbol{x}_*$ is not observed, but $\boldsymbol{x}$ is observed and $\boldsymbol{\eta}$ is measurement error. So

$$\boldsymbol{y} = \boldsymbol{x}\boldsymbol{\beta} + (\boldsymbol{\varepsilon} - \boldsymbol{\eta}\boldsymbol{\beta}) \tag{5.95}$$
$$\tag{5.96}$$

where $\boldsymbol{\varepsilon} - \boldsymbol{\eta}\boldsymbol{\beta}$ is the disturbance.

Now the explanatory variable $\boldsymbol{x}$ and the disturbance are correlated, since both are affected by $\boldsymbol{\eta}$,

$$\mathrm{E}(\boldsymbol{\varepsilon} - \boldsymbol{\eta}\boldsymbol{\beta} \mid \boldsymbol{x}) \neq \boldsymbol{0}. \tag{5.97}$$

So the classic assumptions are not valid for this case of errors in variables.

5.3.2 Properties of the OLS estimator

If we assume $\boldsymbol{y}$ has been generated by the classical linear model (5.91), we can study the stochastic properties of the OLS estimator $\hat{\boldsymbol{\beta}}$. As $\boldsymbol{y}$ is random,

$$\mathrm{E}(\boldsymbol{y}) = \mathrm{E}(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{X}\boldsymbol{\beta}, \tag{5.98}$$
$$\mathrm{Var}(\boldsymbol{y}) = \mathrm{Var}(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \boldsymbol{I}_n. \tag{5.99}$$

As $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$ is a function of $\boldsymbol{y}$, we find

$$\mathrm{E}(\hat{\boldsymbol{\beta}}) = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\,\mathrm{E}\,\boldsymbol{y} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{\beta}. \tag{5.100}$$

So, $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$. Furthermore,

$$\begin{aligned}
\mathrm{Var}(\hat{\boldsymbol{\beta}}) &= \mathrm{Var}\left\{(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}\right\} && (5.101)\\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\,\mathrm{Var}(\boldsymbol{y})\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} && (5.102)\\
&= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'(\sigma^2\boldsymbol{I}_n)\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} && (5.103)\\
&= \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}. && (5.104)
\end{aligned}$$

Summarizing we have

$$\mathrm{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \tag{5.105}$$
$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}. \tag{5.106}$$

However, both the expectation and the covariance matrix are unknown.

Although both $\boldsymbol{\beta}$ and σ^2 are unknown, we can consider estimators of $\boldsymbol{\beta}$ and σ^2 that are "close" to $\boldsymbol{\beta}$ and σ^2, respectively, in some sense. Estimators are statistics that can be computed based on the observations $(\boldsymbol{y}, \boldsymbol{X})$ in the sample.

For example, we might consider using the OLS solution $\hat{\boldsymbol{\beta}}$ as an estimator for $\boldsymbol{\beta}$. It has a good property: $\mathrm{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$. So the expected difference between the estimator and the parameter, which is called the bias of the estimator, $\mathrm{E}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$, is zero.

Now suppose element (i, i) of $\sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$ is small. Then $\mathrm{Var}(\hat{\beta}_i) = \mathrm{E}(\hat{\beta}_i - \beta_i)^2$ is small and we might expect $\hat{\beta}_i$ to be indeed close to β_i.

Remark 4
Asymptotics
Sometimes we consider what happens if the sample size n would increase: $n \to \infty$. As $\boldsymbol{X}'\boldsymbol{X}$ is a summation of n psd matrices

$$\boldsymbol{X}'\boldsymbol{X} = \sum_{i=1}^{n} \boldsymbol{X}'\boldsymbol{e}_i\boldsymbol{e}_i'\boldsymbol{X}, \tag{5.107}$$

we might have

$$\boldsymbol{e}_j'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{e}_j = \sum_{i=1}^{n} (\boldsymbol{e}_i'\boldsymbol{X}\boldsymbol{e}_j)^2 \to \infty \quad \text{as} \quad n \to \infty. \tag{5.108}$$

If it holds for $j = 1, \ldots, p$, such that $\boldsymbol{e}_j'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{e}_j \to 0$, we have $\mathrm{Var}(\hat{\boldsymbol{\beta}}) \to \mathbf{O}$, and $\hat{\boldsymbol{\beta}}$ is said to be consistent, since the bias is zero and the variance matrix converges to zero. So $\boldsymbol{\beta}$ is an unbiased estimator of $\boldsymbol{\beta}$, and if the rows of $\boldsymbol{X}$ behave in a regular way, i.e. $(\boldsymbol{X}'\boldsymbol{X})^{-1} \to \mathbf{0}$, then it is consistent as well.

Suppose there is another estimator $\tilde{\boldsymbol{\beta}}$ that is unbiased as well, $\mathrm{E}(\tilde{\boldsymbol{\beta}}) = \boldsymbol{\beta}$, with a covariance matrix that is smaller than the one found for OLS, i.e.

$$\mathrm{Var}(\tilde{\boldsymbol{\beta}}) \leq \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}. \tag{5.109}$$

In that case we might prefer the alternative estimator $\tilde{\boldsymbol{\beta}}$. We would prefer an estimator that is unbiased with covariance matrix that is smallest in the p.s.d. sense. Such an unbiased estimator would be called efficient.

However, an important theorem says the OLS estimator is Best (efficient) within the class of Linear Unbiased Estimators (BLUE). A linear estimator is linear combination of the elements of $\boldsymbol{y}$, given by $\boldsymbol{C}\boldsymbol{y}$, where $\boldsymbol{C}$ is a fixed, nonrandom matrix.

Theorem 1 Gauss-Markov
Under assumptions $(i) - (iii)$, the OLS estimator is BLUE.

Proof
Let $\tilde{\boldsymbol{\beta}} = \boldsymbol{C}\boldsymbol{y}$ such that $\mathrm{E}(\tilde{\boldsymbol{\beta}}) = \boldsymbol{\beta}$, then $\boldsymbol{C}\,\mathrm{E}(\boldsymbol{y}) = \boldsymbol{C}\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{\beta}$. The validity of this equality should not depend on the particular value of $\boldsymbol{\beta}$. So $\boldsymbol{C}$ should satisfy $\boldsymbol{C}\boldsymbol{X} = \boldsymbol{I}_p$. We find

$$\begin{aligned}
\mathrm{Var}(\tilde{\boldsymbol{\beta}}) &= \boldsymbol{C}\,\mathrm{Var}(\boldsymbol{y})\boldsymbol{C}' && (5.110)\\
&= \sigma^2\boldsymbol{C}\boldsymbol{C}' && (5.111)\\
&\geq \sigma^2\boldsymbol{C}\boldsymbol{P}_{\boldsymbol{X}}\boldsymbol{C}' && (5.112)\\
&= \sigma^2\boldsymbol{C}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{C}' && (5.113)\\
&= \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1} && (5.114)\\
&= \mathrm{Var}(\hat{\boldsymbol{\beta}}). && (5.115)
\end{aligned}$$

So $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$ is BLUE indeed. q.e.d.

5.3.3 Estimation of σ^2

The OLS estimator $\hat{\boldsymbol{\beta}}$ has some desirable properties under the assumptions (i), (ii), and (iii) of the classic stochastic linear model (5.91). We have

$$\begin{aligned}
\mathrm{E}(\hat{\boldsymbol{\beta}}) &= \boldsymbol{\beta}, && (5.116)\\
\mathrm{Var}(\hat{\boldsymbol{\beta}}) &= \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}. && (5.117)
\end{aligned}$$

However, σ^2 is unknown. Would there be an unbiased estimator of σ^2?

Indeed, there is. First compute

$$\begin{aligned}
\mathrm{E}(\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}) &= \mathrm{E}(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})'(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}) && (5.118)\\
&= \mathrm{E}\,\boldsymbol{y}'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y} && (5.119)\\
&= \mathrm{E}\,\boldsymbol{\varepsilon}'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{\varepsilon} && (5.120)\\
&= \mathrm{E}\,\mathrm{tr}\,\{\boldsymbol{\varepsilon}'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{\varepsilon}\} && (5.121)\\
&= \mathrm{E}\,\{\mathrm{tr}(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\} && (5.122)\\
&= \mathrm{tr}\,\{(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{X}})\,\mathrm{E}\,\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\} && (5.123)\\
&= \sigma^2\,\mathrm{tr}(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{X}}) && (5.124)\\
&= \sigma^2(n-p). && (5.125)
\end{aligned}$$

So if we define

$$s^2 = \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n-p} \qquad (5.126)$$

we have an unbiased estimator for σ^2 given by s^2, i.e.

$$\mathrm{E}(s^2) = \sigma^2. \qquad (5.127)$$

The statistic s is called the *standard error of the regression.*

We also have unbiased estimators for the elements of $\mathrm{Var}(\hat{\boldsymbol{\beta}})$ given by

$$\begin{aligned}
\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) &= s^2(\boldsymbol{X}'\boldsymbol{X})^{-1} && (5.128)\\
\mathrm{E}\left\{\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}})\right\} &= \mathrm{Var}(\hat{\boldsymbol{\beta}}). && (5.129)
\end{aligned}$$

5.3.4 Summary of the main results

- Model

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{5.130}$$

where

(i) $\boldsymbol{X} : n \times p$, f.c.r. nonrandom,
(ii) $\mathrm{E}(\boldsymbol{\varepsilon}) = \boldsymbol{0}$,
(iii) $\mathrm{Var}(\boldsymbol{\varepsilon}) = \mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}') = \sigma^2 \boldsymbol{I}_n$.

- OLS

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} \tag{5.131}$$

$$\hat{\boldsymbol{\varepsilon}} = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}, \tag{5.132}$$

$$s^2 = \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n-p}, \tag{5.133}$$

$$\mathrm{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \tag{5.134}$$

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}, \tag{5.135}$$

$$\mathrm{E}(s^2) = \sigma^2. \tag{5.136}$$

Furthermore,

$$\boldsymbol{X}'\hat{\boldsymbol{\varepsilon}} = \boldsymbol{0} \tag{5.137}$$

$$\boldsymbol{y}'\boldsymbol{y} = \hat{\boldsymbol{\beta}}\boldsymbol{X}'\boldsymbol{X}\hat{\boldsymbol{\beta}} + \hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}. \tag{5.138}$$

- Gauss-Markov: $\hat{\boldsymbol{\beta}}$ is BLUE.

- Simple regression: $\boldsymbol{y} = \boldsymbol{\iota}\boldsymbol{\beta}_1 + \boldsymbol{x}\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$

$$\hat{\beta}_2 = \frac{s_{xy}}{s_x^2} = r\frac{s_y}{s_x} \tag{5.139}$$

$$\hat{\beta}_1 = \bar{y} - \bar{x}\hat{\beta}_2 \tag{5.140}$$

$$s_y^2 = s_x^2\hat{\beta}_2^2 + s_{\hat{\varepsilon}}^2 = r^2 s_y^2 + s_{\hat{\varepsilon}}^2, \tag{5.141}$$

i.e.

$$\frac{\boldsymbol{y}'(\boldsymbol{I} - \boldsymbol{P}_\iota)\boldsymbol{y}}{n-1} = \frac{\boldsymbol{x}'(\boldsymbol{I} - \boldsymbol{P}_\iota)\boldsymbol{x}}{n-1}\hat{\beta}_2^2 + \frac{\hat{\boldsymbol{\varepsilon}}'(\boldsymbol{I} - \boldsymbol{P}_\iota)\hat{\boldsymbol{\varepsilon}}}{n-1}. \tag{5.142}$$

5.4 Sum-of-squares decomposition (5.2.4)

	Total	Regression	Residual
Sum of squares (SS)	$\boldsymbol{y}'\boldsymbol{y}$	$\boldsymbol{y}'\boldsymbol{P}_{\boldsymbol{X}}\boldsymbol{y}$	$\boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y}$
Degrees of freedom (df)	n	p	$n-p$
Mean squares (MS)	$\frac{\boldsymbol{y}'\boldsymbol{y}}{n}$	$\frac{\boldsymbol{y}'\boldsymbol{P}_{\boldsymbol{X}}\boldsymbol{y}}{p}$	$\frac{\boldsymbol{y}'(\boldsymbol{I}-\boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y}}{n-p}$
Expected mean squares (EMS)	$\sigma^2 + \frac{\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}}{n}$	$\sigma^2 + \frac{\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}}{p}$	σ^2

A comparison of $\boldsymbol{y}'\boldsymbol{P_X}\boldsymbol{y}/p$ and $\boldsymbol{y}'(\boldsymbol{I}-\boldsymbol{P_X})\boldsymbol{y}/(n-p)$ might give information about a hypothesis $\boldsymbol{\beta}=\boldsymbol{0}$. In order to test such a hypothesis additional assumptions about the distribution of $\boldsymbol{\varepsilon}$ are necessary.

5.5 Restricted least squares (5.2.5)

Consider least squares when there exist q linear constraints given by

$$\boldsymbol{H}\boldsymbol{\beta}=\boldsymbol{0}, \tag{5.143}$$

where $\boldsymbol{H}$ is a $q\times p$ of rank q (full row rank), so $q\leq p$.

Remark 5
linearly independent linear restrictions
Each separate row of $\boldsymbol{H}$ describes a linear restriction, $\boldsymbol{e}_i'\boldsymbol{H}\boldsymbol{\beta}=0$. An example would be

- $\beta_2=0$*: a row of* $\boldsymbol{H}$ *would equal* $\boldsymbol{e}_2'=(0,1,0,\ldots,0)$*, since* $\beta_2=\boldsymbol{e}_2'\beta=0$*;*
- $\beta_2=2\beta_3$*: then* $\boldsymbol{e}_2'-2\boldsymbol{e}_3'$ *would be a row of* $\boldsymbol{H}$*;*
- *If both the restrictions* $\beta_2=0$ *and* $\beta_2=2\beta_3$ *are applied, then the additional restriction* $\beta_3=0$ *would provide no further information. In fact, the matrix with rows* $\boldsymbol{e}_2'$*,* $\boldsymbol{e}_2'-2\boldsymbol{e}_3'$ *and* $\boldsymbol{e}_3'$ *would not have full row rank.*

Therefore, we require $\boldsymbol{H}$ *to have full row rank.*

The RLS solution $\hat{\boldsymbol{\beta}}_{\text{RLS}}$, which is $\hat{\boldsymbol{\beta}}_0$ in the book, is given by the value

$$\hat{\boldsymbol{\beta}}_{\text{RLS}}=\underset{\boldsymbol{\beta}:\ \boldsymbol{H}\boldsymbol{\beta}=\boldsymbol{0}}{\arg\min}\quad(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}) \tag{5.144}$$

The solution is found by using a q-vector $\boldsymbol{\alpha}$ of Lagrange multipliers, and minimizing

$$\boldsymbol{f}(\boldsymbol{\alpha},\boldsymbol{\beta})=(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta})+2\boldsymbol{\alpha}'\boldsymbol{H}\boldsymbol{\beta} \tag{5.145}$$

The normal equations are given by

$$\left.\frac{\partial\boldsymbol{f}(\boldsymbol{\alpha},\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}\right|_{(\hat{\alpha},\hat{\boldsymbol{\beta}}_{\text{RLS}})}=-2\boldsymbol{X}'(\boldsymbol{y}-\boldsymbol{X}\hat{\boldsymbol{\beta}}_{\text{RLS}})+2\boldsymbol{H}'\hat{\boldsymbol{\alpha}}=\boldsymbol{0} \tag{5.146}$$

$$\left.\frac{\partial\boldsymbol{f}(\boldsymbol{\alpha},\boldsymbol{\beta})}{\partial\boldsymbol{\alpha}}\right|_{(\hat{\alpha},\hat{\boldsymbol{\beta}}_{\text{RLS}})}=2\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\text{RLS}}=\boldsymbol{0}. \tag{5.147}$$

To solve for $\boldsymbol{\beta}$ and $\boldsymbol{\alpha}$ premultiply the first equation by $(\boldsymbol{X}'\boldsymbol{X})^{-1}$, which gives

$$\hat{\boldsymbol{\beta}}_{\text{RLS}}=\hat{\boldsymbol{\beta}}_{\text{OLS}}-(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\hat{\boldsymbol{\alpha}}, \tag{5.148}$$

where $\hat{\boldsymbol{\beta}}_{\text{OLS}}=(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$. Premultiplying by $\boldsymbol{H}$ then gives

$$\hat{\boldsymbol{\alpha}}=\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1}\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\text{OLS}}, \tag{5.149}$$

so

$$\hat{\boldsymbol{\beta}}_{\text{RLS}}=\hat{\boldsymbol{\beta}}_{\text{OLS}}-(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1}\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\text{OLS}}, \tag{5.150}$$

which is the RLS solution. Notice $\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\text{RLS}}=\boldsymbol{0}$.

Remark 6
Projections for restricted least squares
Notice

$$\boldsymbol{X}\hat{\boldsymbol{\beta}}_{RLS} = \boldsymbol{P_X y} - \boldsymbol{P}_{\{\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\}}\boldsymbol{y} = \left(\boldsymbol{P_X} - \boldsymbol{P}_{\{\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\}}\right)\boldsymbol{y} = \boldsymbol{P}_0\boldsymbol{y} \tag{5.151}$$

where

$$\boldsymbol{P}_{\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1}\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \tag{5.152}$$

is a projection matrix, projecting on $\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'$*, and* $\boldsymbol{P}_0$*, which is defined implicitly, is a projection matrix as well.* $\boldsymbol{P}_0$ *projects on the space spanned by vectors* $\boldsymbol{X\beta}$ *with* $\boldsymbol{H\beta} = \boldsymbol{0}$*.*

Example 3
Example 5.2.9 in the book
Let

$$y = r(x) + \boldsymbol{\varepsilon} \tag{5.153}$$

and

$$r(x) = \begin{cases} \beta_1 + \beta_2 x, & x \leq x_0, \\ \beta_3 + \beta_4 x + \beta_5 x^2, & x > x_0. \end{cases} \tag{5.154}$$

When $x = x_0$ *we want* $\boldsymbol{r}(x)$ *to be continues, i.e.*

$$\beta_1 + \beta_2 x_0 = \beta_3 + \beta_4 x_0 + \beta_5 x_0^2, \tag{5.155}$$

and the first derivative should be continuous as well

$$\beta_2 = \beta_4 + 2\beta_5 x_0. \tag{5.156}$$

The restrictions say

$$\boldsymbol{H\beta} = \begin{pmatrix} 1 & x_0 & -1 & -x_0 & -x_0^2 \\ 0 & 1 & 0 & -1 & -2x_0 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{pmatrix} = \boldsymbol{0}. \tag{5.157}$$

Let the observations be arranged such that $x_i \leq x_0$ *for* $i = 1, \ldots, m$ *and* $x_i > x_0$ *for* $i = m+1, \ldots, n$*. Then the first* m *rows of* $\boldsymbol{X}$ *are given by* $(1, x_i, 0, 0, 0)$*,* $i = 1, \ldots, m$ *and the last* $n - m$ *rows are given by* $(0, 0, 1, x_i, x_i^2)$*,* $i = m+1, \ldots, n$*.*

The RLS solution is simply given by

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \\ \hat{\beta}_4 \\ \hat{\beta}_5 \end{pmatrix}_{RLS} = \hat{\boldsymbol{\beta}}_{OLS} - (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1}\boldsymbol{H}\hat{\boldsymbol{\beta}}_{OLS}. \tag{5.158}$$

The vector $\boldsymbol{y}$ can be decomposed into three orthogonal components.

$$\boldsymbol{y} = \boldsymbol{X}\hat{\boldsymbol{\beta}}_{RLS} + \boldsymbol{X}(\hat{\boldsymbol{\beta}}_{OLS} - \hat{\boldsymbol{\beta}}_{RLS}) + \hat{\boldsymbol{\varepsilon}}_{OLS} = \boldsymbol{X}\hat{\boldsymbol{\beta}}_{RLS} + \hat{\boldsymbol{\varepsilon}}_{RLS} \tag{5.159}$$

where $\boldsymbol{X}\hat{\boldsymbol{\beta}}_{RLS}$, $\boldsymbol{X}(\hat{\boldsymbol{\beta}}_{OLS} - \hat{\boldsymbol{\beta}}_{RLS})$, and $\hat{\boldsymbol{\varepsilon}}_{OLS}$ are orthogonal to each other. So,

$$||\boldsymbol{y}||^2 = ||\boldsymbol{X}\hat{\boldsymbol{\beta}}_{RLS}||^2 + ||\boldsymbol{X}\hat{\boldsymbol{\beta}}_{OLS} - \boldsymbol{X}\hat{\boldsymbol{\beta}}_{RLS}||^2 + ||\hat{\boldsymbol{\varepsilon}}_{OLS}||^2. \tag{5.160}$$

Question 1 *Can you show* (5.160) *is true?*

Remark 7
Affine restrictions
If $\boldsymbol{\beta}$ is subject to restrictions of the form $\boldsymbol{H\beta} = \boldsymbol{h}$, where $\boldsymbol{H}$ is a nonrandom matrix of full row rank, and $\boldsymbol{h}$ is a nonrandom vector, then

$$\hat{\boldsymbol{\beta}}_{RLS} = \hat{\boldsymbol{\beta}}_{OLS} - (\boldsymbol{X}'\boldsymbol{X})\boldsymbol{H}'\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1}(\boldsymbol{H}\hat{\boldsymbol{\beta}} - \boldsymbol{h}). \tag{5.161}$$

The derivation is similar to the case where $\boldsymbol{h} = \boldsymbol{0}$. However, a remark as in Remark 6 would be more difficult to formulate. Therefore, the book omits this case.

5.6 Normal theory

(Section 5.3 in the book)

5.6.1 The model

The classic model (5.91)

$$\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{\varepsilon}, \tag{5.162}$$

(i) $\boldsymbol{X} : n \times p$, f.c.r., nonrandom,

(ii) $\mathrm{E}(\boldsymbol{\varepsilon}) = \boldsymbol{0}$,

(iii) $\mathrm{E}(\boldsymbol{\varepsilon\varepsilon}') = \sigma^2\boldsymbol{I}_n$,

does only specify the first two moments of the disturbances. Under normality we add a fourth assumption, given by

(iv) $\boldsymbol{\varepsilon}$ is multivariate normal.

So, (ii), (iii), and (iv) can be summarized as

$$\boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2\boldsymbol{I}_n). \tag{5.163}$$

It follows that $\boldsymbol{y}$ is normally distributed as well

$$\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{X\beta},\ \sigma^2\boldsymbol{I}_n). \tag{5.164}$$

The density is given by

$$f(\boldsymbol{y}; \boldsymbol{\beta}, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{\frac{-1/2(\boldsymbol{y}-\boldsymbol{X\beta})'(\boldsymbol{y}-\boldsymbol{X\beta})}{\sigma^2}}. \tag{5.165}$$

The log likelihood is given by

$$l(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2}\frac{(\boldsymbol{y}'\boldsymbol{y} - 2\boldsymbol{y}'\boldsymbol{X\beta} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X\beta})}{\sigma^2}. \tag{5.166}$$

(a regular exponential family, with minimal sufficient statistic $(\boldsymbol{y}'\boldsymbol{X}, \boldsymbol{y}'\boldsymbol{y})$ of dimension $p+1$)

5.6.2 ML estimators

The MLE is easily found. As the likelihood increases as $(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta})$ decreases, we find $\hat{\boldsymbol{\beta}}_{\text{OLS}}$ is the maximum likelihood estimator of $\boldsymbol{\beta}$.

The concentrated log likelihood is thus given by

$$l(\hat{\boldsymbol{\beta}}, \sigma^2) = -\frac{n}{2}\ln\sigma^2 - \frac{1}{2}\frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{\sigma^2}, \tag{5.167}$$

so

$$\hat{\sigma}^2_{\text{ML}} = \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n}. \tag{5.168}$$

Question 2 *Can you verify* (5.168)*?*

We find $\hat{\sigma}^2_{\text{ML}}$ is different from the unbiased estimator

$$s^2 = \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n-p}. \tag{5.169}$$

So, $\hat{\sigma}^2_{\text{ML}}$ is biased, but it is consistent, since

$$\text{E}(\hat{\sigma}^2_{\text{ML}}) \to \sigma^2, \qquad \text{Var}(\hat{\sigma}^2_{\text{ML}}) \to 0, \tag{5.170}$$

as $n \to \infty$. The derivation is based on

$$\hat{\sigma}^2_{\text{ML}} = \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n} = \frac{\boldsymbol{\varepsilon}'(\boldsymbol{I}-\boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{\varepsilon}}{n} \sim \frac{\sigma^2}{n}\chi^2_{n-p}. \tag{5.171}$$

So, as $n \to \infty$,

$$\text{E}(\hat{\sigma}^2_{\text{ML}}) = \left(\frac{n-p}{n}\right)\sigma^2 \to \sigma^2 \tag{5.172}$$

$$\text{Var}(\hat{\sigma}^2_{\text{ML}}) = 2(n-p)\frac{\sigma^4}{n^2} \to 0. \tag{5.173}$$

Consequently, the ML estimators $\hat{\boldsymbol{\beta}}_{\text{ML}}$ and $\hat{\sigma}^2_{\text{ML}}$ are consistent estimators of the parameters in $\boldsymbol{\beta}$ and σ^2, respectively.

5.6.3 A test statistic

The normality and the decomposition

$$\boldsymbol{y}'\boldsymbol{y} = \boldsymbol{y}'\boldsymbol{P}_{\boldsymbol{X}}\boldsymbol{y} + \boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y} \tag{5.174}$$

show us

$$\text{SS}_{\text{Tot}} = \boldsymbol{y}'\boldsymbol{y} \sim \sigma^2\chi^2_n(\delta), \tag{5.175}$$

$$\text{SS}_{\text{Regr}} = \boldsymbol{y}'\boldsymbol{P}_{\boldsymbol{X}}\boldsymbol{y} \sim \sigma^2\chi^2_p(\delta), \tag{5.176}$$

$$\text{SS}_{\text{Res}} = \boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y} \sim \sigma^2\chi^2_{n-p}, \tag{5.177}$$

where $\mathrm{SS}_{\mathrm{Regr}}$ and $\mathrm{SS}_{\mathrm{Res}}$ are independent and $\delta = \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}/\sigma^2$. We find

$$\mathrm{E}(\mathrm{MS}_{\mathrm{Tot}}) = \frac{\sigma^2(n+\delta)}{n} = \sigma^2 + \frac{\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}}{n}, \tag{5.178}$$

$$\mathrm{E}(\mathrm{MS}_{\mathrm{Regr}}) = \frac{\sigma^2(p+\delta)}{p} = \sigma^2 + \frac{\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}}{p}, \tag{5.179}$$

$$\mathrm{E}(\mathrm{MS}_{\mathrm{Res}}) = \frac{\sigma^2(n-p)}{n-p} = \sigma^2, \tag{5.180}$$

and under the null hypothesis $\mathrm{H}_0 : \boldsymbol{\beta} = \mathbf{0}$,

$$T = \frac{\boldsymbol{y}'\boldsymbol{P}_{\boldsymbol{X}}\boldsymbol{y}/p}{\boldsymbol{y}'(\boldsymbol{I}-\boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y}/(n-p)} \sim \mathcal{F}_{p,n-p}. \tag{5.181}$$

So T could be used as a test statistic for testing $\mathrm{H}_0 : \boldsymbol{\beta} = \mathbf{0}$ against $\mathrm{H}_1 : \boldsymbol{\beta} \neq \mathbf{0}$.

Under the alternative, the numerator has a noncentral distribution with noncentrality parameter δ. So for large values of T we would reject the null.

This test can be based on the likelihood ratio test criterion. To test $\mathrm{H}_0 : \boldsymbol{\beta} = \mathbf{0}$ against $\mathrm{H}_1 : \boldsymbol{\beta} \neq \mathbf{0}$ we can use the Likelihood Ratio Test.

Under the null we have

$$\hat{\sigma}_0^2 = \frac{\hat{\boldsymbol{\varepsilon}}_0'\hat{\boldsymbol{\varepsilon}}_0}{n} = \frac{\boldsymbol{y}'\boldsymbol{y}}{n}, \tag{5.182}$$

and under the alternative

$$\hat{\boldsymbol{\beta}}_1 = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}, \tag{5.183}$$

$$\sigma_1^2 = \frac{\hat{\boldsymbol{\varepsilon}}_1\hat{\boldsymbol{\varepsilon}}_1}{n} = \frac{(\boldsymbol{y}-\boldsymbol{X}\hat{\boldsymbol{\beta}})'(\boldsymbol{y}-\boldsymbol{X}\hat{\boldsymbol{\beta}})}{n}. \tag{5.184}$$

So

$$L(\hat{\boldsymbol{\beta}}_1, \hat{\sigma}_1^2) = (2\pi\hat{\sigma}_1^2)^{-n/2} e^{-1/2\frac{\hat{\boldsymbol{\varepsilon}}_1'\hat{\boldsymbol{\varepsilon}}_1}{\hat{\sigma}_1^2}}, \tag{5.185}$$

and

$$L(\hat{\sigma}_0^2) = (2\pi\hat{\sigma}_0^2)^{-n/2} e^{-1/2\frac{\boldsymbol{y}'\boldsymbol{y}}{\hat{\sigma}_0^2}}. \tag{5.186}$$

The likelihood ratio criterion leads to the test statistic

$$\lambda = \frac{L(\hat{\sigma}_0^2)}{L(\hat{\boldsymbol{\beta}}_1, \hat{\sigma}_1^2)} = \left(\frac{\hat{\sigma}_1^2}{\hat{\sigma}_0^2}\right)^{n/2} = \left(\frac{\hat{\boldsymbol{\varepsilon}}_1'\hat{\boldsymbol{\varepsilon}}_1}{\boldsymbol{y}'\boldsymbol{y}}\right)^{n/2}. \tag{5.187}$$

We find the monotonic transformation

$$\begin{aligned}\frac{(n-p)}{p}\left(\lambda^{-2/n}-1\right) &= \frac{(n-p)}{p}\frac{(\boldsymbol{y}'\boldsymbol{y}-\hat{\boldsymbol{\varepsilon}}_1'\hat{\boldsymbol{\varepsilon}}_1)}{\hat{\boldsymbol{\varepsilon}}_1'\hat{\boldsymbol{\varepsilon}}_1} \\ &= \frac{\boldsymbol{y}\boldsymbol{P}_{\boldsymbol{X}}\boldsymbol{y}/p}{\boldsymbol{y}'(\boldsymbol{I}_n-\boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y}/(n-p)} = \frac{\mathrm{MS}_{\mathrm{Regr}}}{\mathrm{MS}_{\mathrm{Res}}} = T.\end{aligned} \tag{5.188}$$

has an $\mathcal{F}_{p,n-p}$ distribution under the null. We reject if λ is small, or T is large.

5.7 Linear hypotheses

(Section 5.3.3 in the book)

5.7.1 An F test

To test $\mathrm{H}_0 : \boldsymbol{H\beta} = \boldsymbol{0}$ against $\mathrm{H}_1 : \boldsymbol{H\beta} \neq \boldsymbol{0}$, where $\boldsymbol{H}$ is a $q \times p$ matrix with full row rank, we can proceed in a way similar to the special case were $\boldsymbol{H} = \boldsymbol{I}_p$. That is,

$$\lambda = \frac{L(\hat{\boldsymbol{\beta}}_0, \hat{\sigma}_0^2)}{L(\hat{\boldsymbol{\beta}}_1, \hat{\sigma}_1^2)} = \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}_1^2}\right)^{-n/2} = \left(\frac{\boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_0)\boldsymbol{y}}{\boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_1)\boldsymbol{y}}\right)^{-n/2}. \tag{5.189}$$

where

$$\boldsymbol{P}_1 = \boldsymbol{P}_{\boldsymbol{X}} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'. \tag{5.190}$$

$$\boldsymbol{P}_0 = \boldsymbol{P}_{\boldsymbol{X}} - \boldsymbol{P}_{\{\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\}}, \tag{5.191}$$

$$\boldsymbol{P}_{\{\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\}} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1}\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}', \tag{5.192}$$

as in Remark 6.

Question 3 *Can you verify the validity of* (5.189)*?*

Now

$$T = \frac{(n-p)}{q}\left(\lambda^{-2/n} - 1\right) = \frac{\boldsymbol{y}'(\boldsymbol{P}_1 - \boldsymbol{P}_0)\boldsymbol{y}/q}{\boldsymbol{y}(\boldsymbol{I}_n - \boldsymbol{P}_1)\boldsymbol{y}/(n-p)} \tag{5.193}$$

$$= \frac{\boldsymbol{y}'\boldsymbol{P}_{\{\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\}}\boldsymbol{y}/q}{\boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y}/(n-p)} \tag{5.194}$$

$$\sim \mathcal{F}_{q,n-p}, \tag{5.195}$$

under the null $\mathrm{H}_0 :\ \boldsymbol{\beta} = \boldsymbol{0}$.

The decomposition of the sum of squares is based on

	$\boldsymbol{y}$	$=$	$\boldsymbol{P}_0\boldsymbol{y}$	$+$	$(\boldsymbol{P}_1 - \boldsymbol{P}_0)\boldsymbol{y}$	$+$	$(\boldsymbol{I}_n - \boldsymbol{P}_1)\boldsymbol{y}$,
SS	$\lVert\boldsymbol{y}\rVert^2$	$=$	$\lVert\boldsymbol{P}_0\boldsymbol{y}\rVert^2$	$+$	$\lVert(\boldsymbol{P}_1 - \boldsymbol{P}_0)\boldsymbol{y}\rVert^2$	$+$	$\lVert(\boldsymbol{I} - \boldsymbol{P}_1)\boldsymbol{y}\rVert^2$,
df	n		$p-q$		q		$n-p$,
MS	$\sigma^2 + \frac{\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}}{n}$		$\sigma^2 + \frac{\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{P}_0\boldsymbol{X}\boldsymbol{\beta}_0}{p-q}$		$\sigma^2 + \frac{\boldsymbol{\beta}'\boldsymbol{X}'(\boldsymbol{P}_1-\boldsymbol{P}_0)\boldsymbol{X}\boldsymbol{\beta}}{q}$		σ^2

5.7.2 An alternative approach

An alternative approach to the relevant test statistics is as follows.

Consider general linear (affine) restrictions

$$\mathrm{H}_0 :\ \boldsymbol{H\beta} = \boldsymbol{h}, \tag{5.196}$$

$$\mathrm{H}_1 :\ \boldsymbol{H\beta} \neq \boldsymbol{h}, \tag{5.197}$$

where $\boldsymbol{H}$ and $\boldsymbol{h}$ are known. The RLS solution is given by

$$\hat{\boldsymbol{\beta}}_{\mathrm{RLS}} = \hat{\boldsymbol{\beta}}_{\mathrm{OLS}} - (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1}(\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} - \boldsymbol{h}). \tag{5.198}$$

Notice

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} \sim \mathcal{N}\left(\boldsymbol{\beta},\ \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\right), \tag{5.199}$$

so

$$\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\text{OLS}} \sim \mathcal{N}\left(\boldsymbol{H}\boldsymbol{\beta},\ \sigma^2\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right), \tag{5.200}$$

and

$$\sigma^{-1}\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1/2}\boldsymbol{H}(\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta}) \sim \mathcal{N}(\boldsymbol{0}.\ \boldsymbol{I}_q) \tag{5.201}$$

Consequently,

$$\sigma^{-2}(\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})'\boldsymbol{H}'\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}\right\}^{-1}\boldsymbol{H}(\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta}) \sim \chi^2_q. \tag{5.202}$$

The left hand side is a function of $\boldsymbol{X}'\boldsymbol{y}$.

Furthermore,

$$\sigma^{-2}\boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y} \sim \chi^2_{n-p}. \tag{5.203}$$

Now the left hand side is a function of $(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y}$. As $\boldsymbol{X}'\boldsymbol{y}$ and $(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y}$ are uncorrelated, they are also independent (due to joint normality).

Consequently, we have

$$\frac{(\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})'\boldsymbol{H}'\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1}\boldsymbol{H}(\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})/q}{\boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y}/(n-p)} \sim \mathcal{F}_{q,n-p}. \tag{5.204}$$

Under the null, $\text{H}_0 : \boldsymbol{H}\boldsymbol{\beta} = \boldsymbol{h}$, it becomes a statistic and

$$T = \frac{(\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{h})'\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1}(\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{h})/q}{s^2} \sim \mathcal{F}_{q,n-p}. \tag{5.205}$$

So T can serve as a test statistic. For $\boldsymbol{h} = \boldsymbol{0}$ the test statistic is the same as the one found in (5.194).

5.7.3 A t test

The $\mathcal{F}$-test is two sided.

$$\text{H}_0 : \boldsymbol{H}\boldsymbol{\beta} = \boldsymbol{h}, \tag{5.206}$$
$$\text{H}_1 : \boldsymbol{H}\boldsymbol{\beta} \neq \boldsymbol{h}. \tag{5.207}$$

We reject if the test statistic is large. If there is a simple restriction, i.e. when $q = 1$, we can also do a one-sided test. Let

$$\text{H}_0 : \boldsymbol{H}\boldsymbol{\beta} = h, \tag{5.208}$$
$$\text{H}_1 : \boldsymbol{H}\boldsymbol{\beta} > h, \tag{5.209}$$

which is possible since now h is a scalar. If $q = 1$ we have

$$\frac{\boldsymbol{H}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})}{\sqrt{\sigma^2 \boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'}} \sim \mathcal{N}(0,\ 1)\,. \tag{5.210}$$

It is a function of $\boldsymbol{X}'\boldsymbol{y}$, therefore it is independent of s^2. So,

$$\frac{\left(\frac{\boldsymbol{H}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})}{\sqrt{\sigma^2 \boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'}}\right)}{\sqrt{\frac{s^2}{\sigma^2}}} \sim t_{n-p}. \tag{5.211}$$

Under the null a one-sided test can be based on

$$T = \frac{\boldsymbol{H}\hat{\boldsymbol{\beta}} - \boldsymbol{h}}{\sqrt{s^2 \boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'}} \sim t_{n-p}. \tag{5.212}$$

It is called a t test. A two-sided test can be based on $T^2 \sim \mathcal{F}_{1,n-p}$, which amounts to an F test.

In particular, if $H = \boldsymbol{e}_i' = (0, \ldots, 0, 1, 0, \ldots, 0)$ and $h = 0$, we have

$$\mathrm{H}_0 : \beta_i = 0, \tag{5.213}$$
$$\mathrm{H}_1 : \beta_i > 0, \tag{5.214}$$

and the t-value is given by

$$t\text{-value} = \frac{\hat{\beta}_i}{\sqrt{s^2 \boldsymbol{e}_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{e}_i}} \sim t_{n-p} \tag{5.215}$$

These are the numbers one frequently finds in tables in papers.

The statistic $\mathrm{se}(\hat{\beta}_i) = s\sqrt{\boldsymbol{e}_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{e}_i}$ is called the standard error of $\hat{\beta}_i$. The unknown standard deviation of $\hat{\beta}_i$ is given by $\sigma\sqrt{\boldsymbol{e}_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{e}_i'}$. Sometimes papers present standard errors instead of t-values.

Remark 8
Efficiency
Notice that under normality $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}$ are independent. In general, when the assumption of normality might not hold, $\hat{\boldsymbol{\beta}}$ is only uncorrelated with $\hat{\boldsymbol{\varepsilon}}$. They should be uncorrelated, since $\hat{\boldsymbol{\beta}}$ is BLUE and $\mathrm{E}(\hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$. This can be shown as follows.

Let $\hat{\theta}$ be an efficient estimator of the scalar parameter θ, so $\mathrm{E}(\hat{\theta}) = \theta$ and no other unbiased estimator has a smaller variance. Let s be a statistic with $\mathrm{E}(s) = 0$, then $\hat{\theta}$ and s are uncorrelated.

Proof *As $\hat{\theta}$ is efficient, it is unbiased. As $\mathrm{E}(s) = 0$, we have $\widetilde{\theta} = \hat{\theta} + \lambda s$ is unbiased as well, for arbitrary nonrandom λ. The variance of $\widetilde{\theta}$ is given by*

$$\mathrm{Var}(\widetilde{\theta}) = \mathrm{Var}(\hat{\theta}) + \lambda^2\,\mathrm{Var}(s) + 2\lambda\,\mathrm{Cov}(\hat{\theta}, s), \tag{5.216}$$

which is minimal for

$$\lambda = -\frac{\mathrm{Cov}(\hat{\theta}, s)}{\mathrm{Var}(s)} \tag{5.217}$$

As $\hat{\theta}$ is efficient, the minimum should be found for $\lambda = 0$. So, $\mathrm{Cov}(\hat{\theta}, s) = 0$, and $\hat{\theta}$ and s are uncorrelated. *q.e.d.*

5.7.4 Confidence sets

For the F-test of $\mathrm{H}_0 : \boldsymbol{H\beta} = \boldsymbol{h}$ against $\mathrm{H}_1 : \boldsymbol{H\beta} \neq \boldsymbol{h}$, we found under the null

$$T = \frac{(\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} - \boldsymbol{h})' \left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1} (\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} - \boldsymbol{h})/q}{s^2} \sim \mathcal{F}_{q,n-p} \tag{5.218}$$

So if $T \sim \mathcal{F}_{q,n-p}$ and $\mathrm{P}(T > T^*) = \alpha$, we reject the null at significance level α if $T > T^*$.

A $(1-\alpha)100\%$ confidence set of vectors $\boldsymbol{H\beta}$ is given

$$C_{1-\alpha} = \left\{ \boldsymbol{H\beta} \,\middle|\, \frac{(\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} - \boldsymbol{H\beta})' \left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1} (\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} - \boldsymbol{H\beta})/q}{s^2} \leq T^* \right\} \tag{5.219}$$

which is an ellipsoid.

If $\boldsymbol{H} = (0, \cdots, 0, 1, 0, \cdots, 0) = \boldsymbol{e}_i'$, then $q = 1$ and the confidence set is an interval. That is, if $q = 1$, then the test statistic is the square of a test statistic with a t_{n-p} distribution. If $\mathrm{H}_0 : \beta_i = \beta_i^*$ against $\mathrm{H}_1 : \beta_i \neq \beta_i^*$ we have under the null

$$T = \frac{\hat{\beta}_i - \beta_i^*}{\sqrt{s^2 \boldsymbol{e}_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{e}_i}} = \frac{\hat{\beta}_i - \beta_i^*}{se(\hat{\beta}_i)} \sim t_{n-p} \tag{5.220}$$

and

$$T^2 = \frac{(\boldsymbol{e}_i'\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} - \beta_i^*)' \left\{\boldsymbol{e}_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}e_i\right\}^{-1} (\boldsymbol{e}_i'\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} - \beta_i^*)/1}{s^2} \tag{5.221}$$

$$= \frac{(\hat{\beta}_i - \beta_i^*)^2}{s^2 \boldsymbol{e}_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{e}_i} \sim \mathcal{F}_{1,n-p}. \tag{5.222}$$

If $T \sim t_{n-p}$ and $\mathrm{P}(T > t^*) = \alpha/2$ then a $(1-\alpha)100\%$ confidence interval for β_i is given by

$$C_{1-\alpha} = \left\{ \beta_i \,\middle|\, \hat{\beta}_i - \mathrm{se}(\hat{\beta}_i)t^* \leq \beta_i \leq \hat{\beta}_i + \mathrm{se}(\hat{\beta}_i)t^* \right\} \tag{5.223}$$

$$= \left\{ \beta_i \,\middle|\, \frac{(\hat{\beta}_i - \beta_i)' \left\{\boldsymbol{e}_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{e}_i\right\}^{-1} (\hat{\beta}_i - \beta_i)}{s^2} \leq t^{*2} \right\}. \tag{5.224}$$

For a one sided test $H_0 : \beta_i = \beta_i^*$ against $H_1 : \beta_i > \beta_i^*$, say, and $P(T > t^{**}) = \alpha$, we reject the null if

$$\frac{\hat{\beta}_i - \beta_i^*}{\sqrt{s^2 \boldsymbol{e}_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{e}_i}} > t^{**} \tag{5.225}$$

and the confidence set is given by

$$C_{1-\alpha} = \left\{ \beta_i \,\middle|\, \frac{\hat{\beta}_i - \beta_i}{\sqrt{s^2 \boldsymbol{e}_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{e}_i}} \leq t^{**} \right\}. \tag{5.226}$$

5.8 Some important applications

(Section 5.4 in the book)

5.8.1 The regression F test

The classic model (5.162) under normality is often considered when $\boldsymbol{x}_1$ is a vector of ones $\boldsymbol{x}_1 = \boldsymbol{\iota}_n$,

$$\boldsymbol{y} = \boldsymbol{\iota}\beta_1 + \boldsymbol{x}_2\beta_2 + \ldots + \boldsymbol{x}_p\beta_p + \boldsymbol{\varepsilon}. \tag{5.227}$$

We then say the regression has a constant term. Frequently, we then test the hypothesis H_0: $\boldsymbol{\beta}_2 = \boldsymbol{0}$ where $\boldsymbol{\beta}_2 = (\beta_2, \ldots, \beta_p)'$ is a $p-1$ vector containing the slope parameters.

Alternatively, the null is given by $\boldsymbol{H}\boldsymbol{\beta} = \boldsymbol{0}$, where $\boldsymbol{H} = (\boldsymbol{0}, \boldsymbol{I}_{p-1})$, and

$$\boldsymbol{y} = \boldsymbol{\iota}\beta_1 + \boldsymbol{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}, \tag{5.228}$$

where $\boldsymbol{X}_2 = (\boldsymbol{x}_2, \ldots, \boldsymbol{x}_p)$.

The F test for this null is called the *Regression F test*, and the regression-F-test statistic is given, similar to (5.205), by

$$F = \frac{\hat{\boldsymbol{\beta}}_{\text{OLS}}'\boldsymbol{H}'\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1}\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\text{OLS}}/(p-1)}{s^2} \tag{5.229}$$

$$= \frac{\hat{\boldsymbol{\beta}}_{\text{OLS}}'(\boldsymbol{0}, \boldsymbol{I}_{p-1})'\left\{(\boldsymbol{0}, \boldsymbol{I}_{p-1})(\boldsymbol{X}'\boldsymbol{X})^{-1}\begin{pmatrix}\boldsymbol{0}' \\ \boldsymbol{I}_{p-1}\end{pmatrix}\right\}^{-1}(\boldsymbol{0}, \boldsymbol{I}_{p-1})\hat{\boldsymbol{\beta}}_{\text{OLS}}/(p-1)}{s^2} \tag{5.230}$$

$$= \frac{\hat{\boldsymbol{\beta}}_{2,\text{OLS}}'\left\{(\boldsymbol{0}, \boldsymbol{I}_{p-1})(\boldsymbol{X}'\boldsymbol{X})^{-1}\begin{pmatrix}\boldsymbol{0}' \\ \boldsymbol{I}_{p-1}\end{pmatrix}\right\}^{-1}\hat{\boldsymbol{\beta}}_{2,\text{OLS}}/(p-1)}{s^2} \tag{5.231}$$

Under the null $F \sim \mathcal{F}_{p-1,n-p}$.

If $\boldsymbol{\iota}'\boldsymbol{X}_2 = \boldsymbol{0}'$, then

$$(\boldsymbol{0}, \boldsymbol{I}_{p-1})(\boldsymbol{X}'\boldsymbol{X})^{-1}\begin{pmatrix}\boldsymbol{0}' \\ \boldsymbol{I}_{p-1}\end{pmatrix} = (\boldsymbol{0}, \boldsymbol{I}_{p-1})\begin{pmatrix} n & \boldsymbol{0}' \\ \boldsymbol{0} & \boldsymbol{X}_2'\boldsymbol{X}_2\end{pmatrix}^{-1}\begin{pmatrix}\boldsymbol{0}' \\ \boldsymbol{I}_{p-1}\end{pmatrix} \tag{5.232}$$

$$= (\boldsymbol{0}, \boldsymbol{I}_{p-1})\begin{pmatrix} n^{-1} & \boldsymbol{0}' \\ \boldsymbol{0} & (\boldsymbol{X}_2'\boldsymbol{X}_2)^{-1}\end{pmatrix}\begin{pmatrix}\boldsymbol{0}' \\ \boldsymbol{I}_{p-1}\end{pmatrix} \tag{5.233}$$

$$= (\boldsymbol{X}_2'\boldsymbol{X}_2)^{-1}. \tag{5.234}$$

If $\boldsymbol{\iota}'\boldsymbol{X}_2 \neq \mathbf{0}'$, then we may use expressions for the inverse of a partitioned matrix to write

$$(\mathbf{0}, \boldsymbol{I}_{p-1})(\boldsymbol{X}'\boldsymbol{X})^{-1}\begin{pmatrix}\mathbf{0}' \\ \boldsymbol{I}_{p-1}\end{pmatrix} = (\mathbf{0}, \boldsymbol{I}_{p-1})\begin{pmatrix}\boldsymbol{\iota}'\boldsymbol{\iota} & \boldsymbol{\iota}'\boldsymbol{X}_2 \\ \boldsymbol{X}_2'\boldsymbol{\iota} & \boldsymbol{X}_2'\boldsymbol{X}_2\end{pmatrix}^{-1}\begin{pmatrix}\mathbf{0}' \\ \boldsymbol{I}_{p-1}\end{pmatrix} \tag{5.235}$$

$$= \{\boldsymbol{X}_2'\boldsymbol{X}_2 - \boldsymbol{X}_2\boldsymbol{\iota}(\boldsymbol{\iota}'\boldsymbol{\iota})^{-1}\boldsymbol{\iota}'\boldsymbol{X}_2\}^{-1} \tag{5.236}$$

$$= \{\boldsymbol{X}_2'(\boldsymbol{I}_n - \boldsymbol{P}_\iota)\boldsymbol{X}_2\}^{-1}. \tag{5.237}$$

So the regression-F-test statistic (5.231) is given by

$$F = \frac{\hat{\boldsymbol{\beta}}_{2,\text{OLS}}'\boldsymbol{X}_2'(\boldsymbol{I}_n - \boldsymbol{P}_\iota)\boldsymbol{X}_2\hat{\boldsymbol{\beta}}_{2,\text{OLS}}/(p-1)}{\boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y}/(n-p)}. \tag{5.238}$$

Remark 9
Variables in deviation from the sample mean
If the regression contains a constant term, in particular when $\boldsymbol{X} = (\boldsymbol{\iota}, \boldsymbol{X}_2)$, then the regression-$F$-test statistic, and inference concerning $\boldsymbol{\beta}_2$, can be formulated in terms of variables in deviation from the sample means, which are given by

$$\widetilde{\boldsymbol{y}} = (\boldsymbol{I}_n - \boldsymbol{P}_\iota)\boldsymbol{y} = \boldsymbol{y} - \boldsymbol{\iota}\overline{y}, \tag{5.239}$$

$$\widetilde{\boldsymbol{X}}_2 = (\boldsymbol{I}_n - \boldsymbol{P}_\iota)\boldsymbol{X}_2 = \boldsymbol{X}_2 - \boldsymbol{\iota}\overline{\boldsymbol{X}}_2, \tag{5.240}$$

where $\overline{y} = \boldsymbol{\iota}'\boldsymbol{y}/n$ and $\overline{\boldsymbol{X}}_2 = \boldsymbol{\iota}'\boldsymbol{X}_2/n$.

In fact the model (5.228) *can be reformulated as*

$$\widetilde{\boldsymbol{y}} = \boldsymbol{\iota}\widetilde{\beta}_1 + \widetilde{\boldsymbol{X}}_2\boldsymbol{\beta}_2 + \varepsilon, \tag{5.241}$$

where $\widetilde{\beta}_1 = \beta_1 - \overline{y} + \overline{\boldsymbol{X}}_2\boldsymbol{\beta}_2$.

As we know the OLS residuals $\hat{\boldsymbol{\varepsilon}}$ are in deviation from the sample mean, since $\boldsymbol{X}'\hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ and in particular $\boldsymbol{\iota}'\hat{\boldsymbol{\varepsilon}} = 0$, just as $\widetilde{\boldsymbol{y}}$ and $\widetilde{\boldsymbol{X}}_2$, but $\boldsymbol{\iota}$ is not *in deviation from the mean, we find*

$$\widetilde{\boldsymbol{y}} = \boldsymbol{\iota}\hat{\widetilde{\boldsymbol{\beta}}}_{1,OLS} + \widetilde{\boldsymbol{X}}_2\hat{\boldsymbol{\beta}}_{2,OLS} + \hat{\boldsymbol{\varepsilon}} \tag{5.242}$$

$$= \widetilde{\boldsymbol{X}}_2\hat{\boldsymbol{\beta}}_{2,OLS} + \hat{\boldsymbol{\varepsilon}}. \tag{5.243}$$

This implies

$$\hat{\beta}_{1,OLS} = \overline{\boldsymbol{y}} - \overline{\boldsymbol{X}}_2\hat{\boldsymbol{\beta}}_{2,OLS}, \tag{5.244}$$

$$\hat{\boldsymbol{\beta}}_{2,OLS} = (\widetilde{\boldsymbol{X}}_2'\widetilde{\boldsymbol{X}}_2)^{-1}\widetilde{\boldsymbol{X}}_2'\widetilde{\boldsymbol{y}}. \tag{5.245}$$

Furthermore,

$$s^2 = \hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}/(n-p) \tag{5.246}$$

$$= \boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y}/(n-p) \tag{5.247}$$

$$= \widetilde{\boldsymbol{y}}'(\boldsymbol{I}_n - \boldsymbol{P}_{\widetilde{\boldsymbol{X}}_2})\widetilde{\boldsymbol{y}}/(n-p) \tag{5.248}$$

In fact the regression-F-test (5.238) *can be written as*

$$F = \frac{\hat{\boldsymbol{\beta}}'_{2,OLS}\widetilde{\boldsymbol{X}}'_2\widetilde{\boldsymbol{X}}_2\hat{\boldsymbol{\beta}}_{2,OLS}/(p-1)}{\widetilde{\boldsymbol{y}}'(\boldsymbol{I}_n - \boldsymbol{P}_{\widetilde{\boldsymbol{X}}_2})\widetilde{\boldsymbol{y}}/(n-p)} \tag{5.249}$$

$$= \frac{\hat{\boldsymbol{\beta}}'_{2,OLS}\widetilde{\boldsymbol{X}}'_2\widetilde{\boldsymbol{X}}_2\hat{\boldsymbol{\beta}}_{2,OLS}/(p-1)}{(\widetilde{\boldsymbol{y}}'\widetilde{\boldsymbol{y}} - \hat{\boldsymbol{\beta}}'_{2,OLS}\widetilde{\boldsymbol{X}}'_2\widetilde{\boldsymbol{X}}_2\hat{\boldsymbol{\beta}}_{2,OLS})/(n-p)} \tag{5.250}$$

Define

$$R^2 = \frac{\hat{\boldsymbol{\beta}}'_{2,\mathrm{OLS}}\widetilde{\boldsymbol{X}}'_2\widetilde{\boldsymbol{X}}_2\hat{\boldsymbol{\beta}}_{2,\mathrm{OLS}}}{\widetilde{\boldsymbol{y}}'\widetilde{\boldsymbol{y}}} \tag{5.251}$$

$$= \frac{\hat{\boldsymbol{\beta}}_{\mathrm{OLS}}\boldsymbol{X}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}})\boldsymbol{X}\hat{\boldsymbol{\beta}}_{\mathrm{OLS}}}{\boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}})\boldsymbol{y}} \tag{5.252}$$

$$= \frac{s^2_{\hat{y}}}{s^2_y}, \tag{5.253}$$

where $\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}}_{\mathrm{OLS}}$, so $\boldsymbol{y} = \hat{\boldsymbol{y}} + \hat{\boldsymbol{\varepsilon}}$ and $\hat{\boldsymbol{y}}'\hat{\boldsymbol{\varepsilon}} = 0$. This R-squared, or coefficient of determination, is the proportion explained sample variance of the regression. The regression-F-test statistic is given in terms of R^2 by

$$F = \left(\frac{n-p}{p-1}\right)\left(\frac{R^2}{1-R^2}\right). \tag{5.254}$$

For simple regression we find

$$F = (n-2)\left(\frac{r^2}{1-r^2}\right), \tag{5.255}$$

where r^2 is the same proportion as in (5.89).

5.8.2 Two-sample t test

Suppose we gave two samples, an n_1 vector $\boldsymbol{y}_1$ and an n_2 vector $\boldsymbol{y}_2$, drawn independently from two populations such that

$$\boldsymbol{y}_1 \sim \mathcal{N}(\beta_1\boldsymbol{\iota}_{n_1},\ \sigma^2\boldsymbol{I}_{n_1}), \tag{5.256}$$

$$\boldsymbol{y}_2 \sim \mathcal{N}(\beta_2\boldsymbol{\iota}_{n_2},\ \sigma^2\boldsymbol{I}_{n_2}). \tag{5.257}$$

Notice the populations have the same variance. We want to test H_0: $\beta_1 = \beta_2$, against H_0: $\beta_1 \neq \beta_2$.

We might proceed as follows. Let

$$\boldsymbol{y} = \begin{pmatrix}\boldsymbol{y}_1\\ \boldsymbol{y}_2\end{pmatrix} = \begin{pmatrix}\boldsymbol{\iota}_{n1} & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{\iota}_{n_2}\end{pmatrix}\begin{pmatrix}\beta_1\\ \beta_2\end{pmatrix} + \boldsymbol{\varepsilon} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{5.258}$$

$$\boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0},\ \sigma^2\boldsymbol{I}_n), \tag{5.259}$$

where $\boldsymbol{X}$ and $\boldsymbol{\beta}$ have been implictly defined and $n = n_1 + n_2$.

We can now test

$$\mathrm{H}_0 : \boldsymbol{H\beta} = (1, -1)\begin{pmatrix}\beta_1\\ \beta_2\end{pmatrix} = \beta_1 - \beta_2 = 0, \tag{5.260}$$

$$\mathrm{H}_1 : \boldsymbol{H\beta} \neq 0. \tag{5.261}$$

The test statistic is given by (5.205), so

$$T = \frac{\hat{\boldsymbol{\beta}}'_{\text{OLS}}\boldsymbol{H}'\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\}^{-1}\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\text{OLS}}}{s^2} \tag{5.262}$$

$$= \frac{(\hat{\beta}_{1,\text{OLS}} - \hat{\beta}_{2,\text{OLS}})'\left\{(1,-1)(\boldsymbol{X}'\boldsymbol{X})^{-1}\left(\begin{smallmatrix}1\\-1\end{smallmatrix}\right)\right\}^{-1}(\hat{\beta}_{1,\text{OLS}} - \hat{\beta}_{2,\text{OLS}})}{s^2} \tag{5.263}$$

$$= \frac{(\hat{\beta}_{1,\text{OLS}} - \hat{\beta}_{2,\text{OLS}})^2}{s^2(1,-1)(\boldsymbol{X}'\boldsymbol{X})^{-1}\left(\begin{smallmatrix}1\\-1\end{smallmatrix}\right)}. \tag{5.264}$$

As $\boldsymbol{X}'\boldsymbol{X}$ is diagonal, we find

$$\hat{\beta}_1 = (\boldsymbol{\iota}'_{n_1}\boldsymbol{\iota}_{n_1})^{-1}\boldsymbol{\iota}'_{n_1}\boldsymbol{y}_1 = \overline{y}_1, \tag{5.265}$$

$$\hat{\beta}_2 = (\boldsymbol{\iota}'_{n_2}\boldsymbol{\iota}_{n_2})^{-1}\boldsymbol{\iota}'_{n_2}\boldsymbol{y}_2 = \overline{y}_2, \tag{5.266}$$

and

$$(\boldsymbol{X}'\boldsymbol{X})^{-1} = \begin{pmatrix} n_1 & 0\\ 0 & n_2\end{pmatrix}^{-1} = \begin{pmatrix} n_1^{-1} & 0\\ 0 & n_2^{-1}\end{pmatrix}, \tag{5.267}$$

so

$$(1,-1)(\boldsymbol{X}'\boldsymbol{X})^{-1}\begin{pmatrix}1\\-1\end{pmatrix} = n_1^{-1`} + n_2^{-1}. \tag{5.268}$$

As a result we have,

$$s^2 = \frac{\boldsymbol{y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}})\boldsymbol{y}}{n-p} = \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n-p} \tag{5.269}$$

$$= \frac{\hat{\boldsymbol{\varepsilon}}'_1\hat{\boldsymbol{\varepsilon}}_1 + \hat{\boldsymbol{\varepsilon}}'_2\hat{\boldsymbol{\varepsilon}}_2}{n_1 + n_2 - 2} \tag{5.270}$$

$$= \frac{\sum_{i=1}^{n_1}(y_{1,i} - \overline{y}_1)^2 + \sum_{i=1}^{n_2}(y_{2,i} - \overline{y}_2)^2}{n_1 + n_2 - 2} \tag{5.271}$$

$$= \frac{(n_1 - 1)s_{y_1}^2 + (n_2 - 1)s_{y_2}^2}{n_1 + n_2 - 2}. \tag{5.272}$$

So, if we know the sample means, $\overline{y}_1$ and $\overline{y}_1$, and the sample variances, $s_{y_1}^2$ and $s_{y_2}^2$, and the sample sizes, n_1 and n_2, we could compute the F test statistic, which is given by

$$T = \frac{(\overline{y}_1 - \overline{y}_2)^2(n_1 + n_2 - 2)}{(n_1^{-1} + n_2^{-1})\left\{(n_1 - 1)s_{y_1}^2 + (n_2 - 1)s_{y_2}^2\right\}}, \tag{5.273}$$

or

$$t = (\overline{y}_1 - \overline{y}_2)\sqrt{\frac{n_1 + n_2 - 2}{(n_1^{-1} + n_2^{-1})\left\{(n_1 - 1)s_{y_1}^2 + (n_2 - 1)s_{y_2}^2\right\}}}, \tag{5.274}$$

and under the null that the population means are equal, $T \sim \mathcal{F}_{1,n_1+n_2-2}$ and $t \sim t_{n_1+n_2-2}$. The latter statistic could also be used in a one-sided test.

5.8.3 Analysis of variance

One-way ANOVA

Suppose in an experiment we have m treatments, including the control treatment. Suppose the number of subjects is the same each time. This *experimental design* produces a *design matrix*

$$\boldsymbol{X} = \begin{pmatrix} \boldsymbol{\iota}_n & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\iota}_n & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \ddots & & \vdots \\ \vdots & \vdots & & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \boldsymbol{\iota}_n \end{pmatrix}. \tag{5.275}$$

For $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)'$, we have

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{5.276}$$

where we assume

$$\boldsymbol{\varepsilon} \sim \mathcal{N}\left(\mathbf{0},\ \sigma^2 \boldsymbol{I}_{nm}\right). \tag{5.277}$$

To test H_0: $\beta_1 = \beta_2 = \ldots = \beta_m$, we compute

$$\hat{\beta}_{i,\mathrm{OLS}} = \overline{y}_i, \tag{5.278}$$

for $i = 1, \ldots, m$. We find

$$\boldsymbol{X}'\boldsymbol{X} = n\boldsymbol{I}_m. \tag{5.279}$$

Let

$$\boldsymbol{H} = (\boldsymbol{\iota}_{m-1},\ -\boldsymbol{I}_{m-1}), \tag{5.280}$$

then the null is given by $\boldsymbol{H}\boldsymbol{\beta} = \mathbf{0}$, which amounts to $q = m - 1$ restrictions.

Furthermore,

$$\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}' = n^{-1}\boldsymbol{H}\boldsymbol{H}' \tag{5.281}$$

$$= n^{-1}\left(\boldsymbol{\iota}_{m-1},\ -\boldsymbol{I}_{m-1}\right)\begin{pmatrix} \boldsymbol{\iota}'_{m-1} \\ -\boldsymbol{I}_{m-1} \end{pmatrix} \tag{5.282}$$

$$= n^{-1}(\boldsymbol{\iota}_{m-1}\boldsymbol{\iota}'_{m-1} + \boldsymbol{I}_{m-1}) \tag{5.283}$$

and

$$\left\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\right\}^{-1} = n\left(\boldsymbol{I}_{m-1} + \boldsymbol{\iota}_{m-1}\boldsymbol{\iota}'_{m-1}\right)^{-1} \tag{5.284}$$

$$= n\left(\boldsymbol{I}_{m-1} - \frac{\boldsymbol{\iota}_{m-1}\boldsymbol{\iota}'_{m-1}}{1+\boldsymbol{\iota}'_{m-1}\boldsymbol{\iota}_{m-1}}\right) \tag{5.285}$$

$$= n\left(\boldsymbol{I}_{m-1} - \frac{\boldsymbol{\iota}_{m-1}\boldsymbol{\iota}'_{m-1}}{m}\right). \tag{5.286}$$

We also have, for $i = 1, \ldots, m$,

$$s^2 = \frac{\boldsymbol{y}'\left(\boldsymbol{I}_{nm} - \boldsymbol{P}_{\boldsymbol{X}}\right)\boldsymbol{y}}{mn-m} = \frac{(n-1)\sum_{i=1}^m s_{y_i}^2}{m(n-1)} = m^{-1}\sum_{i=1}^m s_{y_i}^2. \tag{5.287}$$

The F-test statistic (5.205) is given by

$$T = \frac{\hat{\boldsymbol{\beta}}'_{\text{OLS}}\boldsymbol{H}'\{\boldsymbol{H}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{H}'\}^{-1}\boldsymbol{H}\hat{\boldsymbol{\beta}}_{\text{OLS}}/(m-1)}{s^2} \tag{5.288}$$

$$\tag{5.289}$$

and under the null it would be distributed as $T \sim \mathcal{F}_{m-1,m(n-1)}$.

It can be rewritten as

$$T = \frac{n(m-1)^{-1}\begin{pmatrix}\overline{y}_2 - \overline{y}_1\\ \vdots \\ \overline{y}_m - \overline{y}_1\end{pmatrix}'\left(\boldsymbol{I}_{m-1} - \frac{\boldsymbol{\iota}_{m-1}\boldsymbol{\iota}'_{m-1}}{m}\right)\begin{pmatrix}\overline{y}_2 - \overline{y}_1\\ \vdots \\ \overline{y}_m - \overline{y}_1\end{pmatrix}}{m^{-1}\sum_{i=1}^m s_{y_i}^2}, \tag{5.290}$$

which amounts to

$$T = n\left\{\frac{(m-1)^{-1}\sum_{i=1}^m(\overline{y}_i - \overline{y})^2}{m^{-1}\sum_{i=1}^m s_{y_i}^2}\right\}. \tag{5.291}$$

$$\tag{5.292}$$

The F-test statistic in an ANOVA context can also be written in terms of quantities known as between-group variance and within-group variance. Define

- between group variance: $\frac{\sum_{i=1}^n\sum_{j=1}^m(\overline{y}_j-\overline{y})^2}{m-1} = n\frac{\sum_{j=1}^m(\overline{y}_j-\overline{y})^2}{m-1}$;
- within-group variance: $\frac{\sum_{i=1}^n\sum_{j=1}^m(y_{ij}-\overline{y}_j)^2}{mn-m}$;
- total (sample) variance: $\frac{\sum_{i=1}^n\sum_{j=1}^m(y_{ij}-\overline{y})^2}{mn-1}$.

In terms of a sum of squares decomposition, or ANOVA table, this would appear as follows.

Component	SS	df	MSE
constant	$nm\overline{y}^2$	1	$nm\overline{y}^2$
treatment	$n\sum_{j=1}^{m}(\overline{y}_j - \overline{y})^2$	$m-1$	beteen-group variance
error	$\sum_{i=1}^{n}\sum_{j=1}^{m}(\overline{y}_{ij} - \overline{y}_j)^2$	$m(n-1)$	within-group variance
Total	$\sum_{i=1}^{n}\sum_{j=1}^{m} y_{ij}^2$	nm	

So, the F-test statistic can be written as

$$T = \frac{\text{between-group variance}}{\text{within-group variance}} \sim \mathcal{F}_{m-1,m(n-1)}. \tag{5.293}$$

More-way ANOVA and analysis of covariance

All kinds of treatments can be combined: soil type, amount of water, hours of sunlight, etc. As long as the effects are additive, the design fits within our classical normal linear model. Significance testing would just amount to testing whether or not the elements of a vector like $\boldsymbol{H\beta}$ equal zero.

For us only the regression formulation is relevant. It is given by $\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{\varepsilon}$, where $\boldsymbol{X}$ has full column rank and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0},\ \sigma^2\boldsymbol{I})$.

This holds as well for *analysis of covariance*, where the design matrix contains not only indicator variables, which have ones and zeros as elements and are sometimes called dummies, but also covariates (to control for the effect they might have on the dependent variable).